

Customer Shopping Behavior Analysis

What drives spending and loyalty?

Python | PostgreSQL | Power BI

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github.com/khtibimane/customer_behavior_analysis

Dataset Overview

3,900 purchases (one per customer) and 18 columns, with no purchase dates.

3,900

Purchases

18

Columns

\$233,081

Total Revenue

Business Questions

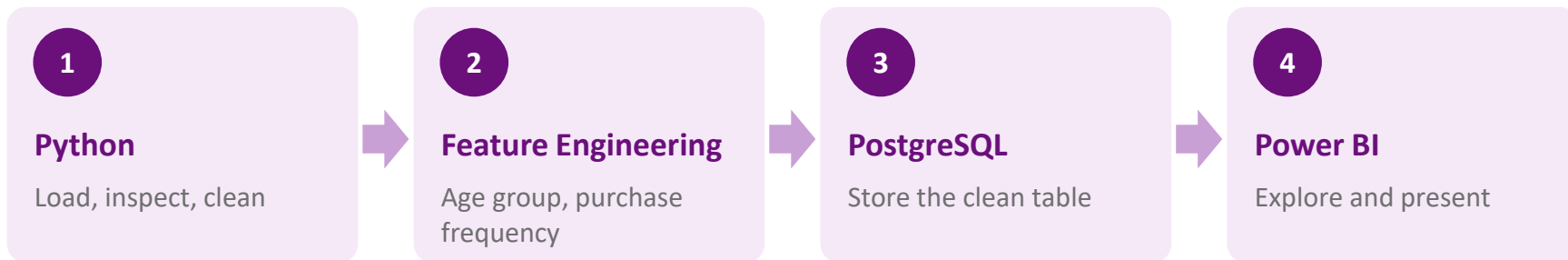
- 01** How loyal are customers, and how many are new?
- 02** Which categories drive revenue, and which products sell most?
- 03** Do subscriptions, discounts or shipping affect spend?
- 04** Do age or gender make a difference?

Data Notes

- 37 missing review ratings (about 1%)
- No purchase dates, so no trends over time

Data Preparation Process

Cleaned in Python, queried in PostgreSQL, and presented in an interactive Power BI dashboard.



Key Cleaning Decisions

- Filled 37 missing ratings with each category's median
- Renamed columns to snake_case for SQL readability
- Dropped promo code: it matched the discount column in every row

Features Created

age_group

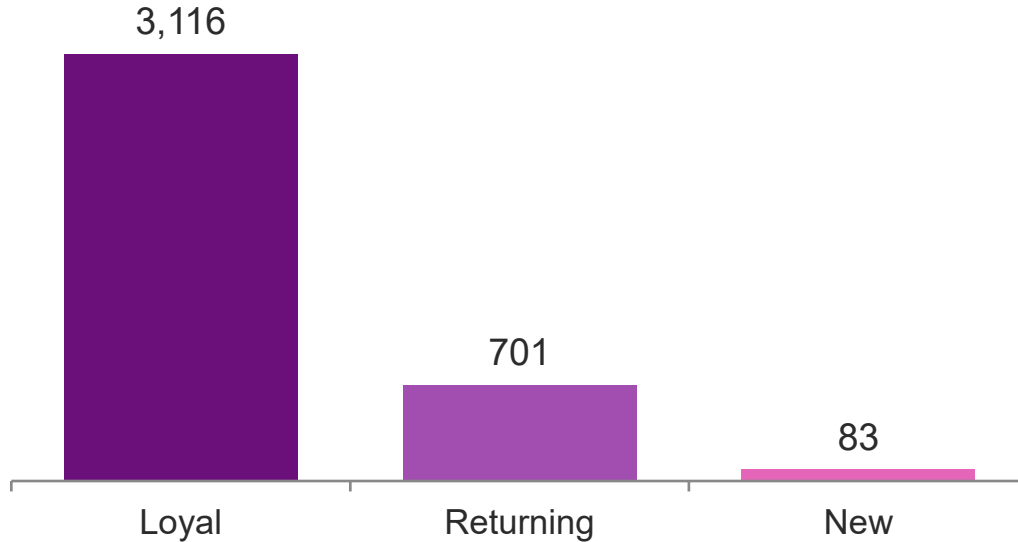
Four age bands using `pd.qcut()`

purchase_frequency_days

Weekly, Monthly, Quarterly as days

Customer Segmentation

Most customers are Loyal; very few are New.



New = 1 previous purchase | Returning = 2 to 10 | Loyal = 11 or more

80%

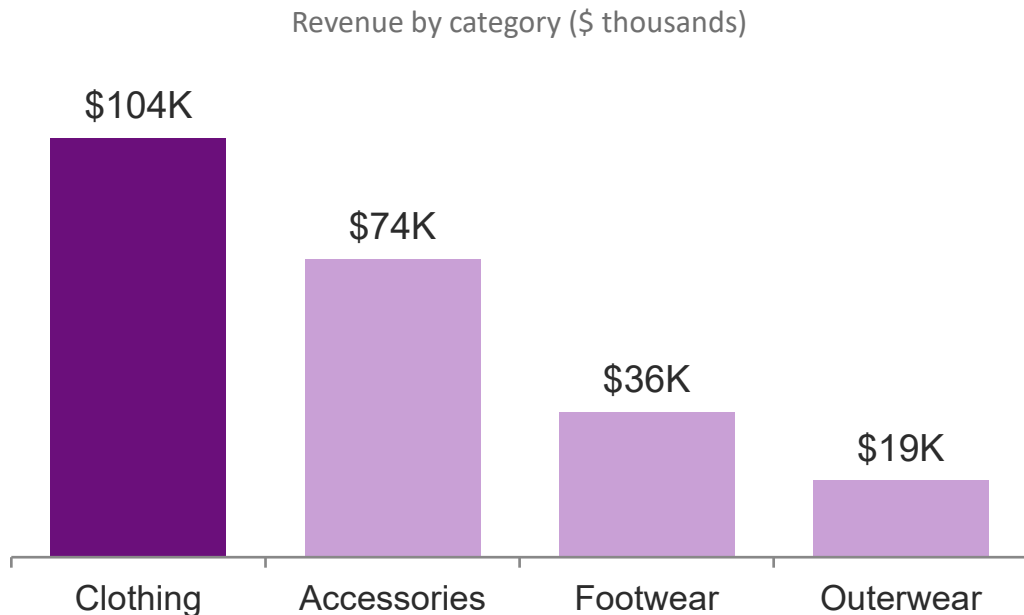
of customers are Loyal (3,116)

83

New customers: the weak spot

Revenue by Category

Clothing brings in the most revenue, about 45% of \$233K.



45%

of revenue comes from Clothing

Most purchased product

Accessories: **Jewelry (171 orders)**

Clothing: **Blouse (171 orders)**

Footwear: **Sandals (160 orders)**

Outerwear: **Jacket (163 orders)**

Top products are close: 145 to 171 orders

Subscriptions, Discounts and Shipping

Subscribing isn't increasing spending, and 43% of purchases used a discount.

Discount usage

43%

of purchases used a discount

1,677 of 3,900

Subscription

\$59.49 vs \$59.87

Average spend per purchase,
subscribers vs non-subscribers

Shipping

\$60.48 vs \$58.46

Express vs Standard, a gap of
about \$2

Most discounted

~50%

Hats (50.00%) and sneakers
(49.66%) are discounted most;
the top five are all close to half

Above-average spenders

839

Customers used a discount and
still spent above the \$59.76
average

Age and Gender

Age barely changes spend; men bring in most of the revenue.

Age

\$59.07 to \$60.45

Average spend across the four age groups.
Young Adults bring in the most revenue (\$62,143).

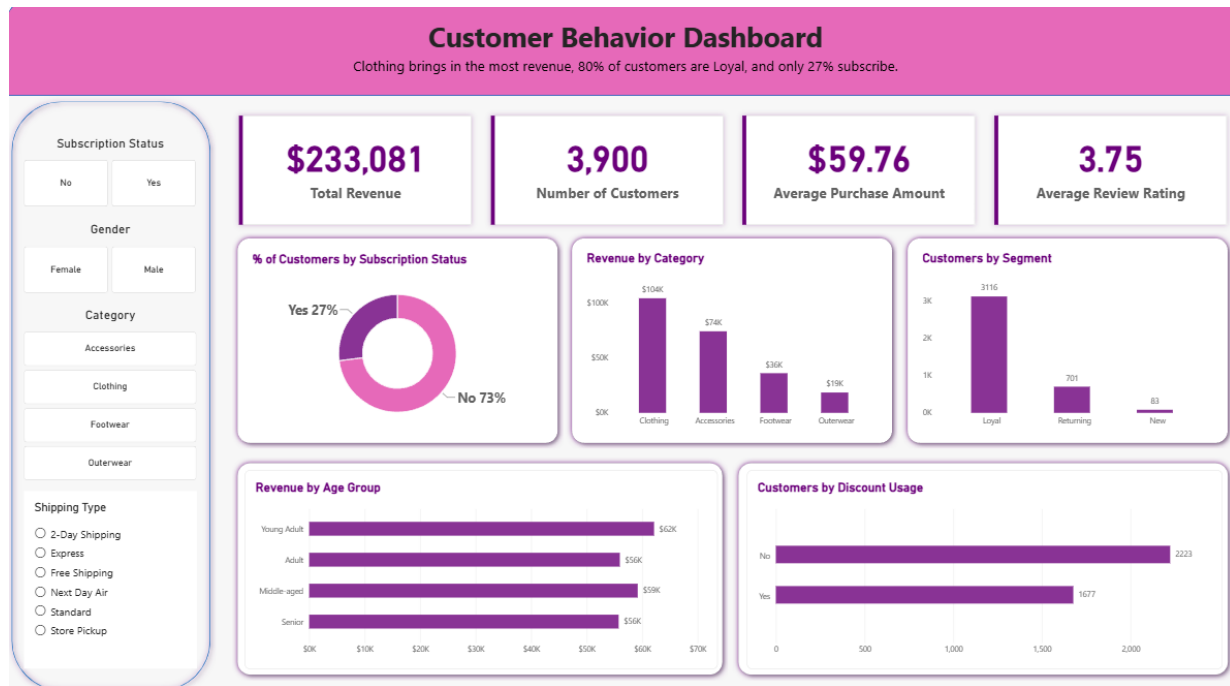
Gender

68% of revenue

Male customers bring in about 68% of revenue (\$157,890 vs \$75,191), roughly twice as much as female customers.

Power BI Dashboard

Interactive, connected to the PostgreSQL customer table.



Filters

Subscription

Gender

Category

Shipping Type

Recommendations

Ideas to test, not proven conclusions.

1

Grow New Customers

Test an acquisition incentive on a first purchase.

START HERE

New-customer volume

2

Rethink Subscription Value

Test benefits designed to encourage larger purchases.

TRACK

Subscriber vs non-subscriber spend

3

Test Discounts Before Extending

Reduce the discount on selected products and monitor demand.

TRACK

Units sold and revenue

4

Build on Leading Categories

Feature top sellers and investigate Outerwear.

TRACK

Category revenue

5

Segment by Behavior

Compare behavioral segments with age-based targeting.

TRACK

Campaign results by segment

Limitations

These constraints shape what the results can and cannot show.

No Date Variable

Trends, growth and retention over time can't be analyzed.

Segment Cutoffs

New, Returning and Loyal thresholds are analyst-defined.

Unequal Group Sizes

Repeat-buyer subscription comparison: 3,476 repeat buyers vs 424 other customers.

No Cost Data

The impact of discounts on profit can't be estimated.

Correlation, Not Causation

Relationships do not prove cause, and no significance tests were run.

Single Dataset

Findings may not generalize beyond these 3,900 purchases.

Full project: github.com/khtibimane/customer_behavior_analysis